Expert Parallelism, from scratch

A custom `expert_parallel` module I built on top of `picotron`

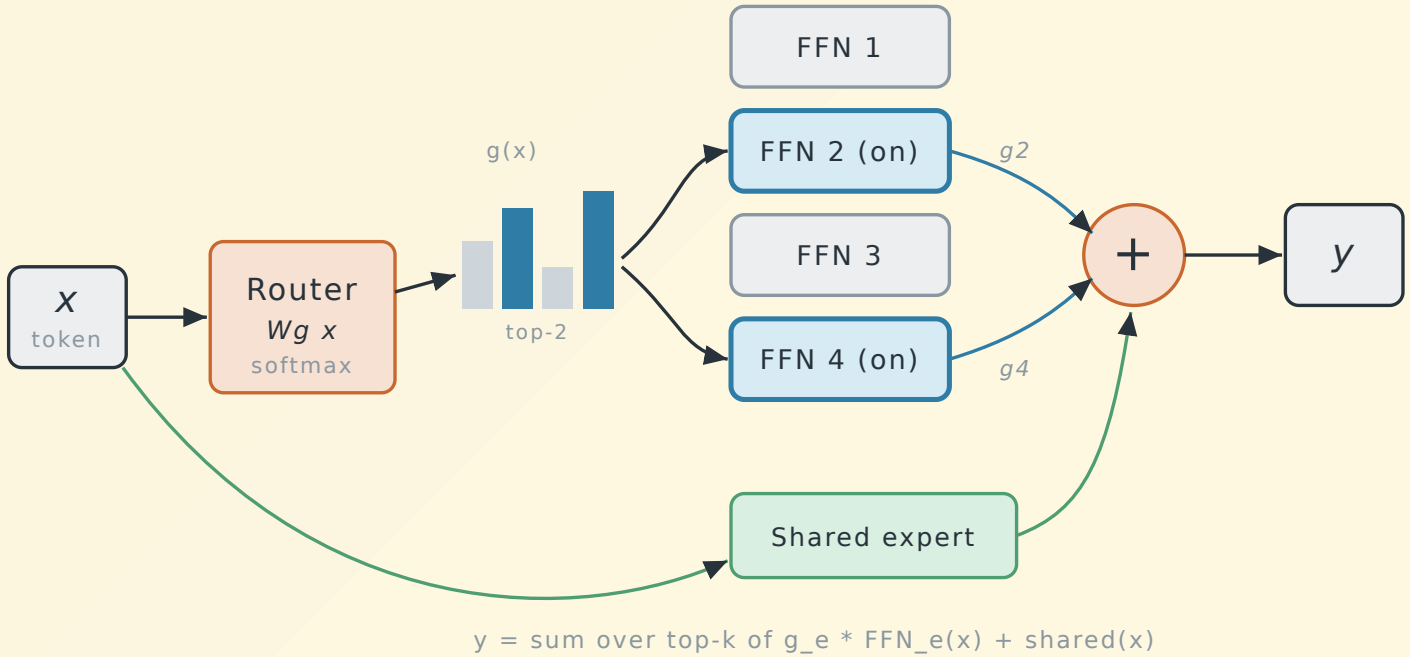
From a naive all-to-all to DeepEP, tiled pipelines, FP8 & LatentMoE

extending picotron (DP·PP·CP·TP) with a 5th axis — expert parallelism — and a stack of MoE comm optimizations
deep dive: autograd · CUDA streams · Triton kernels · the bandwidth lesson

What we are optimizing

A **Mixture-of-Experts** FFN: a router picks the top-`k` of `E` experts per token.

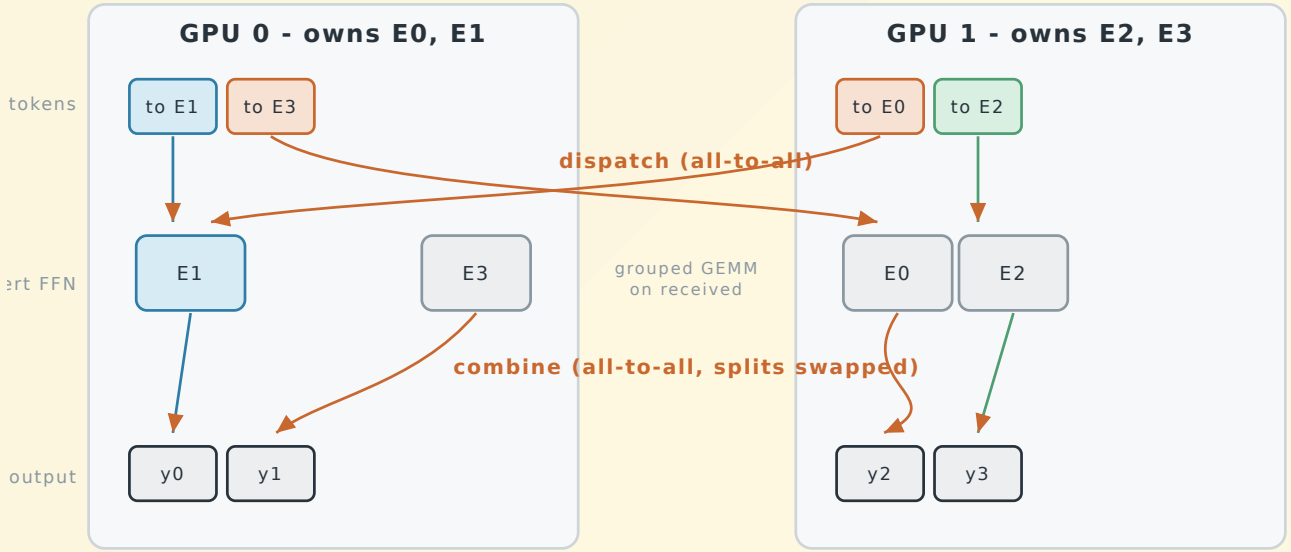
$$y_t = \sum_{e \in \text{top-}k(t)} g_{t,e}\,\text{FFN}_e(x_t) \;+\; \text{shared}(x_t), \qquad g_{t,e} = \text{softmax}(W_g x_t)_e$$



Only `k` of `E` experts run per token (sparse). The router is trained; the shared expert is

EP: tokens travel to their expert and back

Expert Parallelism: shard the `E` experts across ranks — rank `r` owns `[r·L, (r+1)·L)`, `L = E/ep_size`.
A token routed to a remote expert needs **two all-to-alls**: dispatch out, combine back.



EP: the 5th axis I added to picotron

Stock picotron is `DP × PP × CP × TP` (4D). I extended its process-group manager to a 5D grid `DP × PP × CP × EP × TP` and added the MoE layer that uses it.

Tensor	Sharded over	Grad sync
Expert weights `FFN_e`	**EP**	none (each rank owns distinct experts)
Router `W_g`, shared expert, norms	replicated	all-reduce over `cp_dp`

Design constraints I kept:

- **Readable** `torch.distributed` collectives (picotron spirit)
- **Bit-exact** across every optimization path (tested vs the naive baseline, `grad_diff ≤ 5e-7`)
- Every knob is **optional** and safe to leave on (auto-fallback)

The challenge: a local op becomes a network round-trip

Without EP, the expert FFN is **pure local compute**. EP spreads experts across GPUs, so every token must be **sent to its expert and sent back** — two all-to-alls wrapped around the compute.

That round-trip is what makes EP hard:

- **It blocks.** Experts can't start until tokens arrive; the layer can't end until results come back. The GPU sits **idle** during both all-to-alls.
- **It gets worse at scale.** More GPUs → traffic leaves the fast NVLink island for slower links, while each GPU's share of compute shrinks. Comm starts to dominate.
- **Backward pays it twice.** The gradient pass repeats both all-to-alls, and plain `autograd` runs them one-after-another — it won't overlap them for you.

→ So we have two levers: make the waiting **invisible** (overlap) or send **less data** (volume).

The two families of optimization

1. Hide the comm (overlap)

Run useful compute *while* the all-to-all is in flight.

- shared-expert overlap (side stream)
- tiled pipeline (fwd **and** bwd)
- DeepEP kernels (near-zero SM)

→ helps **only** when comm is a real fraction of the step.

2. Send fewer bytes (volume)

Shrink the payload itself.

- FP8 (E4M3) dispatch
- LatentMoE (route/compute in a small latent dim)

→ helps even with **nothing left to hide behind**.

Orthogonal. They stack: `latent + fp8` cuts bytes 8×, and overlap hides whatever remains.

Step 0 — the naive baseline

`dispatch → experts → combine`, blocking, fully differentiable.

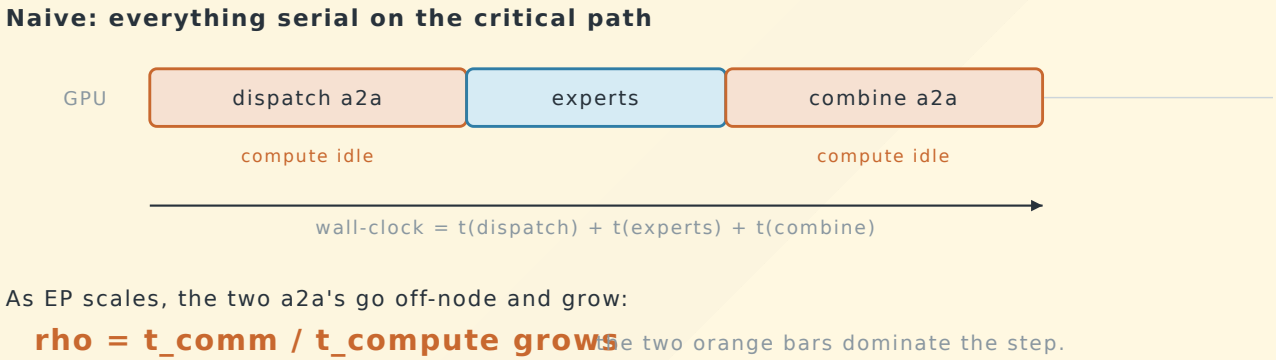
```
def _moe_naive(self, tokens, topk_idx, topk_weights):
    routed, expert_idx, w = self._expand_routed(tokens, topk_idx, topk_weights)
    order, inv, in_list, out_list, local_expert = self._dest_plan(expert_idx)
    recv          = all_to_all(routed[order], out_list, in_list, group)   # dispatch
    recv_out      = self._run_local_experts(recv, local_expert)           # grouped GEMM
    combined      = all_to_all(recv_out, in_list, out_list, group)        # combine
    y = self._combine_topk(combined[inv], w, num_tokens)                  # weight + sum top-k
    return self._finalize(y, shared)
```

The all-to-all itself is one autograd `Function` — **backward of an a2a is the reverse a2a**:

```
class _AllToAll(torch.autograd.Function):
    def forward(ctx, group, x, out_splits, in_splits): ...   # all_to_all_single(out, x, out, in)
    def backward(ctx, g):  ...                                # all_to_all_single(gin, g, in, out)  ← swapped
```

The question that decides everything

How much of the step is communication? $\rho = t_{\text{comm}} / t_{\text{compute}}$



- $\rho \gg 1$ (cross-node IB/RoCE, large EP) → **overlap is gold** · $\rho \ll 1$ (NVLink island) → **nothing to hide**

We *measure* ρ on three fabrics and watch the same code flip from **1.6× win** to **no-op**.

Opt 1 — shared-expert overlap (the orthogonal one)

DeepSeek-style models add an **always-on shared expert** = pure local compute, **no comm**.

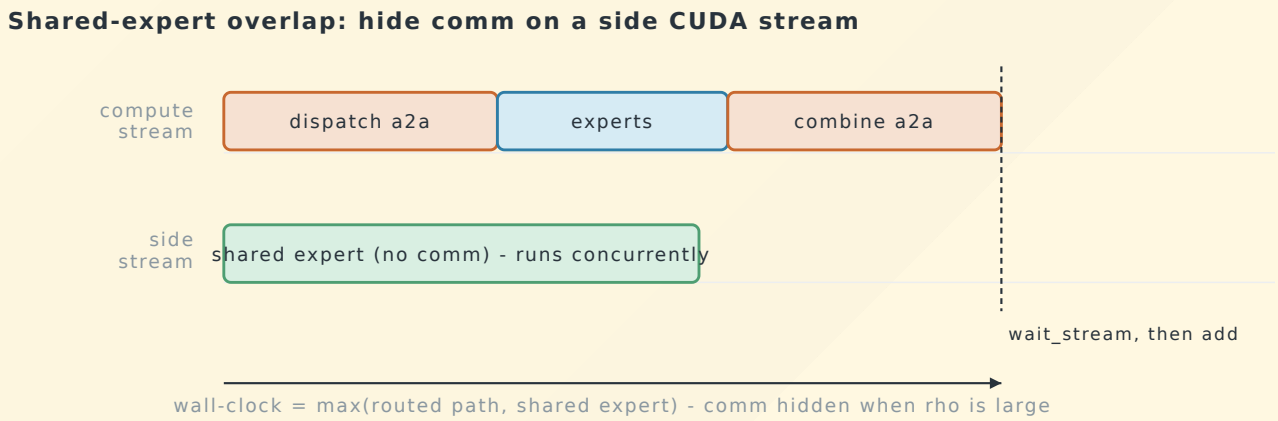
Idea: run it on a **side CUDA stream** so it overlaps the routed path's all-to-all.

```
def _shared_start(self, tokens):
    if not (self.ep_overlap and self.ep_world_size > 1 and tokens.is_cuda):
        return self.shared_expert(tokens), None          # serial (CPU/gloo, 1 rank)
    self._shared_stream.wait_stream(torch.cuda.current_stream())  # tokens ready
    with torch.cuda.stream(self._shared_stream):
        out = self.shared_expert(tokens)                 # runs concurrently with dispatch/combine
    return out, self._shared_stream

def _finalize(self, y, shared):
    out, stream = shared
    if stream is not None:
        torch.cuda.current_stream().wait_stream(stream)  # join before the add
    return y + out
```

Key design choice: this is *not* a 4th path. It composes with **naive / tiled / deepep** alike.

Opt 1 — deep notes (streams + autograd)

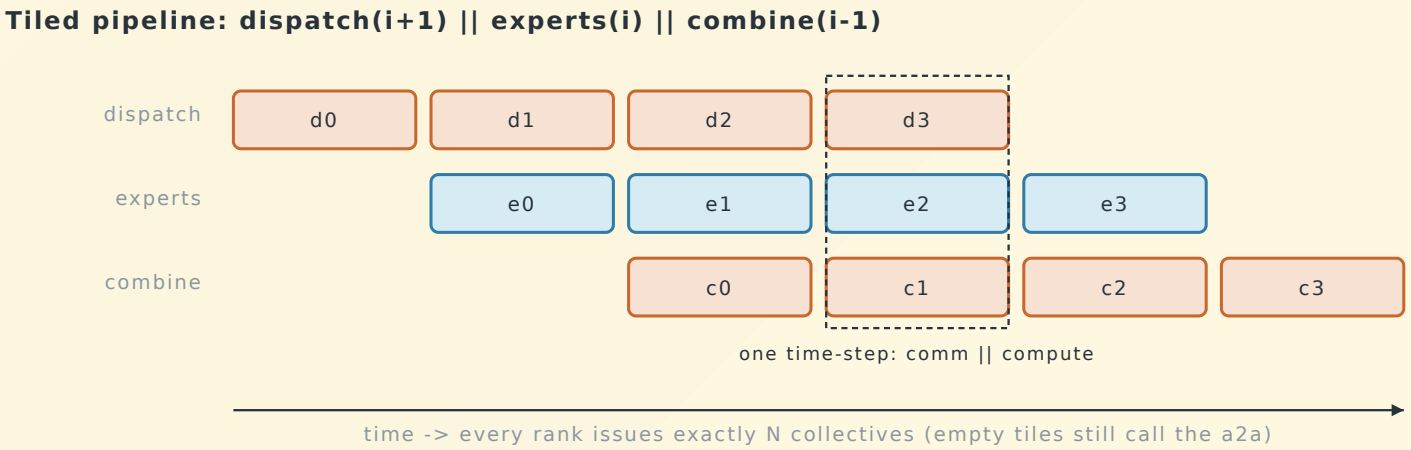


- **Correctness:** `wait_stream` on entry + before the add; reads of `tokens` on two streams don't conflict.
- **Backward:** autograd replays the shared-expert backward on the side stream (one benign `AccumulateGrad` stream-mismatch warning) — verified **bit-exact** (`out_diff = grad_diff = 0` on 2× A100 NCCL).
- **Best on DeepEP:** its comm uses ~0 SMs, so the GPU is free for the shared FFN.

`num_shared_experts = N` merges into one MLP of width `N·intermediate` (= sum of N experts, one GEMM).

Opt 2 — tiled pipeline (forward)

Split routed tokens into `N` tiles; software-pipeline the three stages (MegaScale-MoE):




```
recv, recv_work, recv_le, le_work = dispatch(plans[0])
for i in range(n):
    nxt = dispatch(plans[i+1]) if i+1 < n else None   # prefetch next tile's a2a
    recv_work.wait()                                   # ... while this tile computes
    out_i = _ffn_forward(recv_sorted, counts, gate_w, up_w, down_w)
    combined, combine_work = all_to_all_no_grad_async(out_i, ...)   # combine in flight
```

Every rank issues exactly `N` collectives (empty tiles still call the zero-sized a2a) → **lockstep**.

Opt 2 — tiled backward (the hard part)

> MegaScale-MoE's lesson, re-discovered the hard way:
> **do not implement the MoE backward with `torch.autograd`.**

A first version calling `torch.autograd.grad` twice per tile regressed the full step to **0.87×**.

Why: autograd schedules backward nodes **serially** — you cannot make it interleave one tile's reverse-a2a with another tile's compute. **You must own the backward schedule.**

Megatron's independence trick: from the same upstream grad, the FFN backward produces `d_input` and `d_weight` **independently**:

```
per tile:  dgrad → launch reverse-dispatch a2a(d_input)  ‖  compute wgrad
           combine_rev(i+1) is prefetched to cover tile i's recompute
```

→ one recompute pass, no double graph traversal, comm hidden behind `wgrad`.

Opt 2 — the Triton megakernel

Production-style fused expert FFN (vLLM / MegaBlocks pattern):

- **Grouped GEMM**: fixed `BLOCK_M=64` token blocks, each block belongs to **one** expert (block-aligned tiling). `triton.autotune` over tile sizes / warps / stages.
- gate & up **fused** into one GEMM `w13 = [E, 2I, H]` → Triton `silu_and_mul` → down GEMM.
- routed-weight epilogue fused into the down-GEMM store.

Explicit backward (no autograd):

$$\texttt{dgrad:}\quad dX = dC \cdot W \quad (\text{reuse fwd GEMM, swap } W \text{ strides: } AW^\top \to AW)$$

$$\texttt{wgrad:}\quad dW = C^\top A \quad (\text{dedicated grouped-GEMM kernel})$$

```
dx_sorted, a_c, di_c = megakernel.fused_moe_dgrad(recv, counts, gate_w, up_w, down_w, d_out)
wgrads = lambda: megakernel.fused_moe_wgrad(recv, counts, a_c, di_c, d_out)  # deferred → overlap
```

Opt 3 — DeepEP backend

Swap our `torch.distributed` a2a for **DeepEP's hand-written CUDA kernels** (Hopper SM90+).

- Saturates NVLink/RDMA at **near-zero SM occupancy** → frees SMs for the GEMM (its real win is the cross-node RDMA path, not raw 2-GPU bandwidth).
- We wrap the **classic intranode** `Buffer` (NVLink P2P), *not* the V2 `ElasticBuffer` — the latter needs NCCL GIN, which aborts on single-node clusters (e.g. Modal H100): *"NCCL GIN is unavailable"*.

```
recv_x, handle, recv_idx, recv_w, _ = deepep_backend.dispatch(group, latent, topk_idx, topk_w, E)
expert_out = self._deepep_experts(recv_x, recv_idx, recv_w)   # grouped FFN + gate weights
y = deepep_backend.combine(group, expert_out, handle)         # sum per-rank contributions
```

`dispatch` and `combine` are **linear transposes** ⇒ autograd backward of one **is** the other. Gate weights ride a non-differentiable transport (router grad flows the normal way).

Always safe: on A100 (sm_80) `deepep_available()` is False → transparent fallback to torch.

Switching families — reduce *volume*

Overlap hides the a2a. But if $\rho \ll 1$, or you're already hiding everything, the next lever is **fewer bytes on the wire**.

FP8 dispatch
`2 B → ~1 B` per element.
Comm-bound win.

LatentMoE
route/compute in dim `l ≪ d`.
Cuts comm **and** FLOPs.

These are *architecture/precision* changes to the payload, independent of *how* it's moved — so they stack with each other **and** with all three overlap paths.

Opt 4 — FP8 (E4M3) dispatch

DeepSeek-V3 recipe: **FP8 dispatch, BF16 combine**.

Per-token scale into E4M3's range (max magnitude **448**):

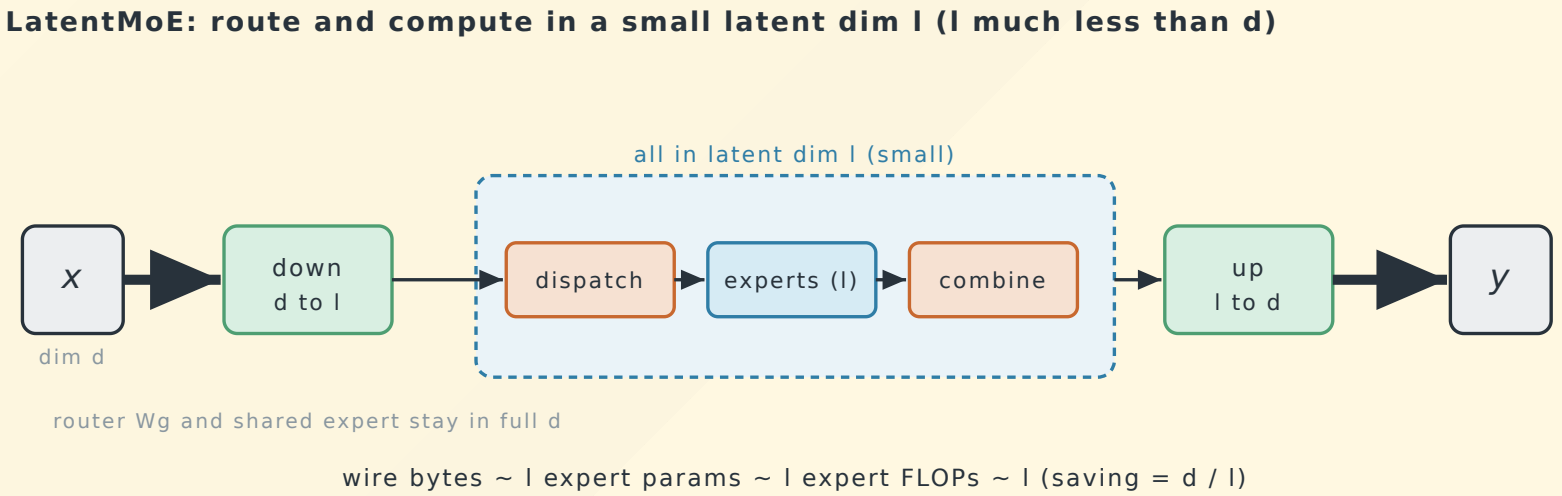
$$s_t = \frac{\max_h |x_{t,h}|}{448}, \qquad \hat{x}_t = \mathrm{clip}\left(\frac{x_t}{s_t}, \pm 448\right) \in \texttt{e4m3}$$

```
q, scale = _per_token_quant_fp8(x)                      # [N,H] fp8, [N] fp32
dist.all_to_all_single(recv_q.view(uint8), q.view(uint8), ...)   # NCCL has no fp8 coll → bitcast
recv = recv_q.to(bf16) * recv_scale[:, None]            # dequant on receiver, GEMM in BF16
```

- Backward is **straight-through** in BF16 (only the forward *activation* dispatch is low precision).
- No FP8 tensor cores needed → **works on A100**.
- Measured: **~2× fewer dispatch bytes**, relative L2 error **0.023**.

Opt 5 — LatentMoE (NVIDIA Nemotron)

Shared **down-projection** `d→l` before routing/dispatch; experts & a2a live in `l`; **up-projection** `l→d` after combine. Router & shared expert stay in full `d`.



- It's an **architecture** change (experts parameterized in `l`) → must be **trained** this way.
- Nemotron reinvests the savings into **more experts / higher top-k** (accuracy-per-FLOP).
- The **only** optimization here that also cuts *compute* ⇒ wins even on fast fabrics.

Stacking the volume reducers

`hidden=4096, latent=1024, inter=4096, tokens=8192`, dispatch payload:

config	wire B / route	vs dense
dense bf16	8192	1.0×
dense + fp8	4100	2.0×
latent (`l=1024`)	2048	4.0×
latent + fp8	**1028**	**8.0×**

The two are multiplicative on bytes. What that buys in wall-clock depends on $\rho \rightarrow$ next.

Results I — PCIe is comm-bound (ρ large)

2× L4, **PCIe**, no NVLink. Single MoE layer (`hidden4096 inter4096 experts8 tokens8192`, dispatch ≈ 134 MB):

path	forward	fwd+bwd
plain	1.00×	1.00×
overlap(shared)	1.10×	1.02×
tiled-3	1.39×	1.25×
tiled-4	**1.62×**	1.27×
tiled-6	1.54×	**1.32×**

End-to-end training step (real `Llama`, 8192 tok/layer): **tiled-4 → 1.28×** (8636 → 11078 tok/s).

→ When the link is slow, hiding the a2a is a **big** win.

Results II — NVLink is compute-bound (ρ tiny)

2× A100, **NV12 ≈ 300 GB/s**. Same code, story **inverts**:

path	fwd (134 MB)	fwd+bwd	fwd (537 MB)
plain	1.00× (21.4 ms)	1.00× (77.4 ms)	1.00× (79.5 ms)
overlap(shared)	1.00×	1.01×	1.01×
tiled-3	0.97×	0.98×	0.98×
tiled-6/8	0.86–0.94×	0.96×	0.91–0.92×

Even a 537 MB payload moves dispatch+combine (~1 GB) in **~3.6 ms** = **~4.5 % of the step**.
Nothing to hide → per-tile overhead **regresses** it. **Default** `ep_num_tiles=1` **on NVLink.**

Results III — controlled proof: it's *bandwidth*, not hardware

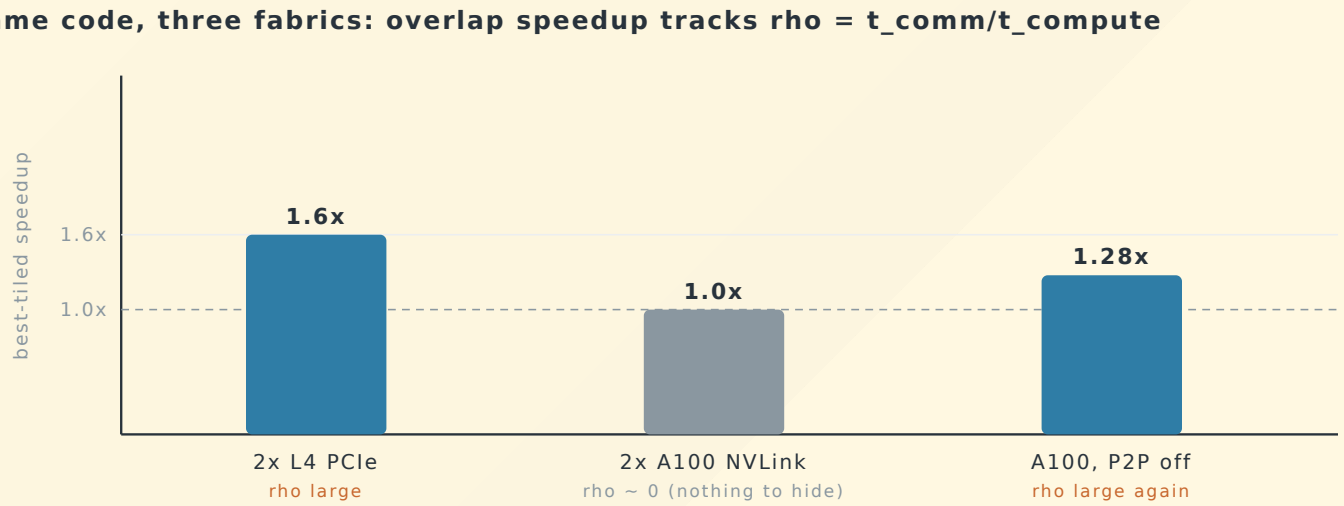
Same A100 box, same config, forward-only. **Only variable: interconnect bandwidth** (`NCCL_P2P_DISABLE=1` forces the slow host path).

interconnect	plain	best tiled	speedup
NVLink on (~300 GB/s)	21.4 ms	22.0 ms	**none** (comm ≈ 4.5%)
NVLink off (slow path)	39.2 ms	**30.5 ms**	**1.28×**
2× L4 PCIe	—	—	**1.3–1.6×**

Flip the bandwidth, the same kernels flip from no-op to 1.28×.

> Reducing `inter` does **not** reproduce this — it leaves comm fast and only exposes fixed per-tile launch overhead (tiled-6 → 0.43×). **Only throttling bandwidth restores the bottleneck.**

The picture: speedup tracks bandwidth, not hardware



Identical code & config. The overlap win appears **only** where comm is a real fraction of the step — exactly the cross-node / large-EP regime where production MoE lives.

Results IV — H100 + DeepEP

Built from source on **2× H100 (NVLink) via Modal** (`modal run modal_run.py`). All correctness checks pass on real Hopper (DeepEP vs torch, fwd **and** bwd).

model	fwd rel err	grad rel err	torch	deepep	speedup
dense	0.0000	0.0000	2.00 ms	1.64 ms	**1.22×**
LatentMoE (`l=512`)	0.0000	0.0000	2.06 ms	1.66 ms	**1.24×**

DeepEP matches torch to bf16 and is ~1.2× faster **even on one NVLink node** (frees SMs, skips our explicit permute). The big wins are at scale / cross-node.

Compress on H100 (fwd): latent **2.41×**, latent+fp8 2.26× — same story as A100.

The big lesson

Overlap (3 paths)

Bandwidth-bound. Pays off iff $\rho = t_{\text{comm}}/t_{\text{compute}}$ is non-trivial:

- cross-node IB/RoCE ✅
- large EP degree ✅
- single-node NVLink ❌

LatentMoE

Compute-bound win too. Cuts FLOPs, so it pays off **even on NVLink** (2.14×).
The one lever that doesn't need a comm bottleneck.

FP8 dispatch

Comm-bound like overlap (+7% off-NVLink, slight regression on NVLink).

Production MoE lives in the **cross-node, large-EP** regime → that's the `NCCL_P2P_DISABLE=1` row, where everything here compounds.

Decision guide

Your situation	Turn on
Single-node NVLink, few tokens/layer	`ep_num_tiles=1`, `ep_overlap` ~free, skip FP8
Single-node NVLink, want speed	**LatentMoE** (cuts compute)
Cross-node / IB / large EP, ≳8k tok/layer	**tiled-N** (+ overlap)
Hopper + want SMs back	**`ep_backend="deepep"`**
Slow fabric, comm-bound	**FP8 dispatch** + LatentMoE (8× bytes)
Always	leave knobs on — all auto-fallback & bit-exact

```
"ep_overlap": true, "ep_num_tiles": 4, "ep_fp8_dispatch": false,
"moe_latent_dim": 1024, "ep_backend": "deepep"
```

Recap — the optimization ladder

1. **Naive** a2a dispatch/combine — differentiable, reverse-a2a backward.
2. **Shared-expert overlap** — side stream, composes with every path.
3. **Tiled pipeline** — fwd ‖; backward needs **explicit kernels** (autograd-twice = 0.87×).
4. **Megakernel** — grouped GEMM; dgrad by stride-swap, dedicated wgrad.
5. **DeepEP** — Hopper kernels, near-zero SM, classic `Buffer`, auto-fallback.
6. **FP8 dispatch** — E4M3 per-token scale, straight-through BF16 backward, 2× bytes.
7. **LatentMoE** — route/compute in `l`, cuts comm **and** FLOPs, stacks to 8× with FP8.

Throughline: measure ρ. Hide comm where it's expensive; cut bytes/FLOPs where it isn't.

Appendix

file map · how to run

File map & how to run

File	Role
`expert_parallel.py`	`MoELayer`: router, sharded experts, path selection, FP8 + LatentMoE
`ep_communications.py`	differentiable / async / **FP8** all-to-all primitives
`tiled_moe.py`	`_TiledMoEFunction`: tiled pipeline, fwd **and** bwd overlap
`megakernel.py`	Triton fused FFN + explicit `dgrad` / `wgrad` grouped GEMM
`deepep_backend.py`	DeepEP dispatch/combine (Hopper, auto-fallback)

```
# correctness (2 ranks, gloo/CPU by default; EP_TEST_BACKEND=nccl for GPU)
torchrun --nproc_per_node 2 tests/test_expert_parallel.py
torchrun --nproc_per_node 2 tests/test_fp8_dispatch.py

# benchmarks (2 GPUs)
torchrun --nproc_per_node 2 tests/bench_ep_overlap.py   --tokens 8192 --tiles 3 4 6 --backward
torchrun --nproc_per_node 2 tests/bench_ep_compress.py  --hidden 4096 --latent 1024 --tokens 8192

# DeepEP / Hopper on cloud GPUs (builds DeepEP from source)
modal run modal_run.py
```